

Aziz Mohammad

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EDUCATION

University of Illinois Chicago, Chicago, IL Aug 2025 – May 2027 (Expected)
BS in Mechanical Engineering
GPA – 3.8/4.0 | Technical GPA – 3.86/4.0 | Dean's List
Relevant Coursework: Compressible Flow, Heat Transfer, Fluid Mechanics, Thermodynamics I & II
College of DuPage, Glen Ellyn, IL Aug 2023 – May 2025
FE Mechanical Exam — Planned Winter 2026–2027

WORK EXPERIENCE

Wilo USA, *Systems Engineering Intern*, Rosemont, IL May 2026 – August 2026

- Led the mechanical design and systems integration of a modular commercial HVAC hydronic pump package platform, standardizing **20+** system configurations to replace one-off custom engineering with a scalable product line.
- Developed large-scale SolidWorks assemblies and 2D general arrangement drawings, integrating grooved piping, insulation, electrical conduit, dual-pump arrangements, VFDs, expansion vessels, buffer tanks, air/dirt separators, and single-point power systems.
- Translated customer and application requirements into manufacturable system configurations while coordinating with mechanical, electrical, sales, and project stakeholders through design reviews and project meetings.
- Built configuration matrices, automated costing tools, and structured BOM datasets in Excel, standardizing components across DIN and U.S. hardware standards to reduce repetitive engineering work and accelerate quotation workflows.

University of Illinois Chicago, *Research Assistant at Intelligent Multiphase Systems Lab* March 2026 – Present

- Developing a precision flow-boiling experimental test platform for investigating multiphase heat transfer and interfacial transport phenomena relevant to high-performance electronics cooling.
- Designed and iterated SolidWorks hardware for thermal-fluid experiments, integrating fluid routing, mechanical interfaces, instrumentation, and manufacturability; used FDM/SLA prototyping with Rigid 10K resin to reduce development time by **20%** and improve assembly consistency by **12%**.
- Integrated and validated experimental instrumentation including flow measurement, DAQ hardware, sensors, signal wiring, and power connections, supporting calibration, troubleshooting, flow-system validation, and repeatable data acquisition.

CLUBS, ACTIVITIES & LEADERSHIP

Society of Automotive Engineers (SAE), *Member* Aug 2025 – May 2026

- Optimized a custom Formula SAE push bar using SolidWorks and ANSYS Workbench, reducing component weight by **20%** while maintaining a **2.0** factor of safety under expected loading conditions.

American Society of Mechanical Engineers (ASME), *Member* Aug 2025 – Present

Muslim Student Association (MSA), *Member/Volunteer* Aug 2023 – Present

- Coordinated 10+ community service and outreach events with teams of 20+ volunteers, supporting event execution, volunteer coordination, and community engagement.

SKILLS

- **Software & Analysis:** SolidWorks, Creo, AutoCAD, ANSYS Workbench, MATLAB, Python, Excel, Google Workspace, Microsoft Office; FEA, engineering data analysis, workflow automation
- **Engineering:** SLA/FDM 3D Printing, CNC, Manual Machining, GD&T, DFM, Drawings, BOM Creation